

OpenBuild — Executive Findings

Analytics engineering portfolio · 108K orders · 87K users · 2019–2022 · Medallion architecture · 32 tested transformations

1. Retention is structurally flat AND the loyalty program is making it worse

RESULT

1a. Cohort timing doesn't matter. Month-1 retention sits between 0.8% and 1.4% across all 48 cohorts (Jan 2019 – Dec 2022). No cohort breaks the band — including the COVID-era peak (Jun 2020 = 2,887 new users). Acquisition strategy did not move repurchase behavior.

1b. Loyalty membership at acquisition correlates with 3.6× worse retention. Loyalty members retain at 0.45% in month 1 vs. 1.63% for non-loyalty (averaged across 48 cohorts). Deficit is front-loaded; converges by month 9. The gap emerged in 2020: 2019 loyalty was nearly on par (1.35% vs. 1.55%); 2020 onward shows 4–6× deficit. Mechanism: loyalty disproportionately recruits AirPods buyers (58.3% of loyalty first-purchases vs. 36.6% non-loyalty) and underweights replenishables (5.7% Charging Cable Pack vs. 34.0% non-loyalty). AOVs equal (\$239 vs. \$257) — *not* a discount effect.

IMPLICATION

Marketing won't fix retention; product mix is the lever. The loyalty program currently amplifies the catalog's retention problem by acquiring into one-and-done categories. Recruitment mechanics need to steer loyalty signups toward replenishables.

CAUSES

Cause	Weight	Status
Durable goods have low repurchase frequency (1.23 orders/user across 4 years)	High	Tested
Catalog skews to one-time purchases (only 1 of 8 products is replenishable)	High	Tested
Loyalty program scaled 11.6% (2019) → 55.8% (2022), recruiting into one-and-done categories	High	Tested
Limited cross-sell mechanics in checkout	Medium	Hypothesis

Source models: *mart_cohort_retention*, *mart_loyalty_retention*, *dim_users*, *fct_orders* · See Appendix A1, A1b

2. Laptops drive 2.5–4.3× the company refund rate

RESULT

Refund baseline: 4.97%. Top offenders all laptops: ThinkPad × CA (21.3%), ThinkPad × GB (16.8%), MacBook Air × CA (16.3%). MacBook Air × US: largest dollar leak, \$365K refunded.

IMPLICATION

Refund-reduction initiatives should target laptops first. Geographic variation suggests fulfillment or returns-policy investigation by country.

CAUSES

Cause	Weight	Status
High-AOV products carry higher absolute return risk	High	Tested
Spec / expectation mismatch in laptops vs. accessories	High	Partially tested
Country-level returns policy or fulfillment variance	Medium	Partially tested
Localization issues (pricing, positioning)	Medium	Hypothesis

Source models: *mart_refund_metrics*, *fct_orders*, *dim_country* · See Appendix A2

3. Mobile is structurally a low-AOV channel

RESULT

Website = 96.8% of lifetime revenue. Mobile generates 17% of orders but only 3% of revenue. Mobile share moved from 2.95% (2019) to 3.93% (2022) — statistically real, business-trivially small.

IMPLICATION

Mobile investment must target AOV, not order volume. Doubling mobile order volume only adds ~3 percentage points to total revenue at current AOV.

CAUSES

Cause	Weight	Status
Mobile users skew toward accessory purchases	High	Tested
Catalog product mix is desktop-shaped (laptops, monitors)	High	Tested
Mobile UX friction at high-value checkout	Medium	Hypothesis
Desktop preference for large / considered purchases	Low–Medium	Hypothesis

Source models: *mart_channel_revenue*, *fct_orders*, *dim_platform* · See Appendix A3

Full analysis, SQL, and tests: github.com/joyce92200/ecommerce-analytics-engineering

Appendix — Methodology

Each finding's metric definition, SQL logic, and caveats. All numbers reproducible from the repository above.

A1 — Cohort retention (timing)

Metric. Period-active retention: % of a cohort placing at least one order during month N after acquisition. Not "made a 2nd purchase ever" (different metric).

Cohort definition. cohort_month = DATE_TRUNC('month', MIN(purchase_ts)) per user.

```
retention_pct = active_users / cohort_size × 100
active_users = COUNT(DISTINCT user_id) per (cohort_month, months_since_acquisition)
cohort_size = active_users at months_since_acquisition = 0
```

Caveats. Cohorts after Dec 2021 right-censored. 3 orders (0.003%) with unparseable timestamps excluded.

A1b — Loyalty retention

Metric. Same retention definition as A1, but split by loyalty status at first purchase. Grain: (cohort_month × loyalty_status × months_since_acquisition).

Anti-confounder. Uses *loyalty_at_first_purchase*, not current loyalty status. This avoids reverse causality where users who retained → joined loyalty later → look like good retainers. Snapshot of loyalty membership at acquisition is the only valid measure for cohort comparison.

```
FROM dim_users u JOIN fct_orders f ON f.user_id = u.user_id
GROUP BY u.first_purchase_month, u.loyalty_at_first_purchase, f.months_since_acquisition
(active_users / cohort_size) per group → retention_pct
```

Sample sizes. Non-loyalty: 48,512 users acquired across 48 cohorts. Loyalty: 39,113 users acquired. Both segments large enough for the gap to be real signal, not statistical noise.

Mechanism investigation. First-purchase product mix and AOV computed from fct_orders WHERE is_first_purchase = 1, joined to dim_users on user_id. AOV difference (\$239 vs \$257 = 7%) is small relative to the 3.6× retention gap, ruling out discount-driven selection as a primary mechanism.

Caveats. "Loyalty" is a binary 0/1 in source data; granularity (tier, points balance, signup channel) is not available. Causes for the 2020 structural break (program eligibility change vs. marketing campaign vs. checkout default) require operational data not in this dataset.

A2 — Refund metrics

Metric. Two metrics at (product × country) grain: *refund_rate_pct* = refunded orders / total × 100; *refunded_revenue_usd* = SUM(usd_price WHERE is_refunded = 1).

Refund definition. Order is refunded if refund_ts is non-null. Partial vs. full refunds not distinguished in source.

```
WHERE orders >= 100
ORDER BY refund_rate_pct DESC
LIMIT 10
```

Caveats. 100-order minimum is a statistical-reliability threshold. 4 orders with NULL country_code retained as a distinct segment. 18 orders with non-standard codes (EU, AP) mapped to Unclassified for referential integrity.

A3 — Channel revenue

Metric. Net revenue (gross minus refunded) by (purchase_month × purchase_platform) with within-month share %.

```
net_revenue_usd = SUM(usd_price WHERE is_refunded = 0)
share_of_month_pct = net_revenue / SUM(net_revenue) OVER (PARTITION BY purchase_month) × 100
```

Caveats. Channel attribution is platform-only (website vs. mobile app). Marketing channel attribution unavailable (upstream column >99% null). AOV comparisons use net_revenue / orders across platforms.

How to verify

Clone the repository, install requirements, open notebooks/01_cohort_exploration.ipynb, and run all cells top-to-bottom. Each finding corresponds to a specific cell whose output matches the numbers reported. The 32 data quality tests in tests/test_models.sql validate every input model.